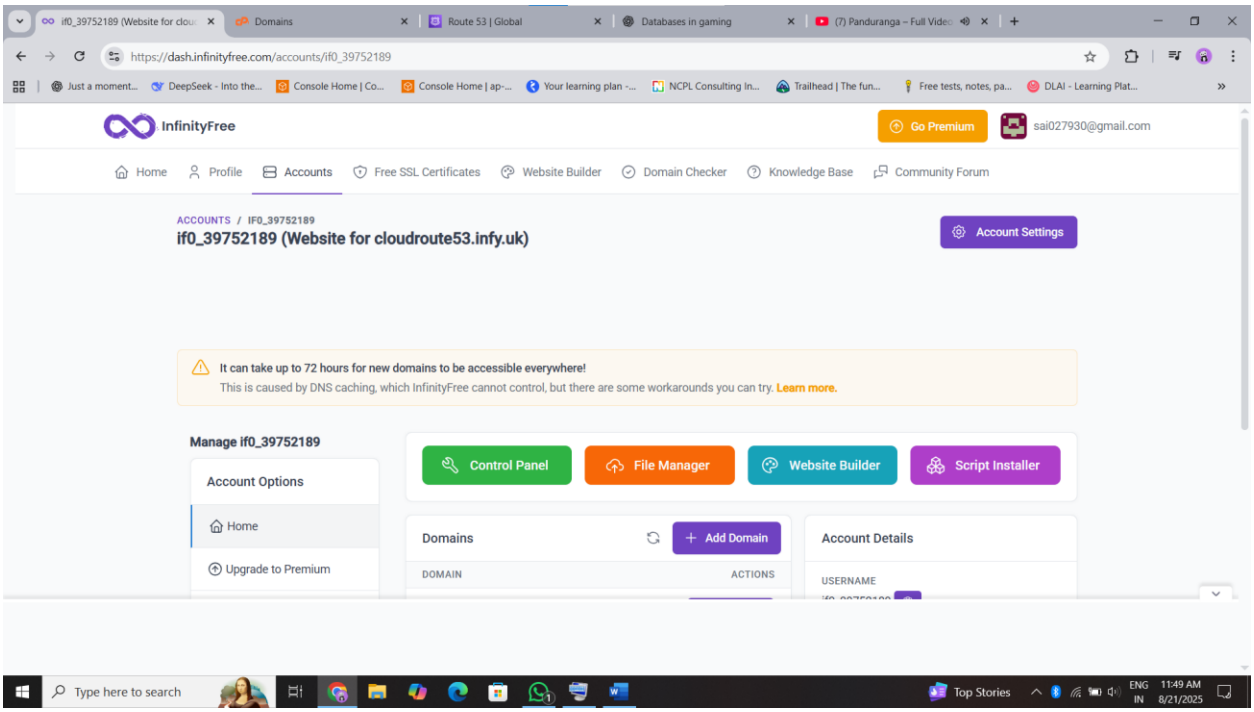
## 🔷 Task 1: Mirror the Demo Exactly

### 1. Get a Domain

- If you already own a domain (GoDaddy, Namecheap, etc.) → use that.
- If not:
  - Buy a domain in Route 53 (around $10–12/year).
  - Or use a cheap registrar (~$1 for first year).
  - Or get a free domain from Freenom & Infinityfree.

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### 2. Create an S3 Bucket for Website Hosting

- Bucket name must exactly match your domain (e.g., www.yourdomain.com).

aws s3 mb s3://www.yourdomain.com --region us-east-1

https://ap-south-1.console.aws.amazon.com/s3/bucket/create?region=ap-south-1&bucketType=general

Amazon S3 > Buckets > Create bucket

**AWS Region**
Asia Pacific (Mumbai) ap-south-1

**Bucket type** Info

- General purpose — Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.
- Directory — Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

**Bucket name** Info

cloudroute53

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). Learn More

**Copy settings from existing bucket - *optional***
Only the bucket settings in the following configuration are copied.

Choose bucket

Format: s3://bucket/prefix

## Object Ownership Info

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

- ACLs disabled (recommended) — All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.
- ACLs enabled — Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

⚠ We recommend disabling ACLs, unless you need to control access for each object individually or to have the object writer own the data they upload. Using a bucket policy instead of ACLs to share data with users outside of your account simplifies permissions management and auditing.

**Object Ownership**

https://ap-south-1.console.aws.amazon.com/s3/buckets?region=ap-south-1&bucketType=general

Amazon S3 > Buckets

✓ **Successfully created bucket "cloudroute53"**
To upload files and folders, or to configure additional bucket settings, choose **View details**.

View details

General purpose buckets | All AWS Regions | Directory buckets

## General purpose buckets (1) Info

Buckets are containers for data stored in S3.

Copy ARN | Empty | Delete | Create bucket

Find buckets by name

| Name | AWS Region | Creation date |
|---|---|---|
| cloudroute53 | Asia Pacific (Mumbai) ap-south-1 | August 20, 2025, 23:03:04 (UTC+05:30) |

**Account snapshot** Info — View dashboard
Updated daily
Storage Lens provides visibility into storage usage and activity trends.

**External access summary - *new*** Info
Updated daily
External access findings help you identify bucket permissions that allow public access or access from other AWS accounts.

https://ap-south-1.console.aws.amazon.com/s3/upload/cloudroute53?region=ap-south-1&bucketType=general

Amazon S3 > Buckets > cloudroute53 > Upload

## Upload Info

Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. Learn more

Drag and drop files and folders you want to upload here, or choose **Add files** or **Add folder**.

**Files and folders** (139 total, 1.4 MB)

All files and folders in this table will be uploaded.

Remove | Add files | Add folder

Find by name

1 2 3 4 5 6 7 ... 14

| Name | Folder | Type | Size |
|---|---|---|---|
| _grid.scss | scss/bootstrap/scss/mixins/ | - | 3.9 KB |
| _image.scss | scss/bootstrap/scss/mixins/ | - | 395.0 B |
| _list-group.scss | scss/bootstrap/scss/mixins/ | - | 509.0 B |
| _lists.scss | scss/bootstrap/scss/mixins/ | - | 168.0 B |
| _pagination.scss | scss/bootstrap/scss/mixins/ | - | 741.0 B |
| _reset-text.scss | scss/bootstrap/scss/mixins/ | - | 495.0 B |
| _resize.scss | scss/bootstrap/scss/mixins/ | - | 202.0 B |
| _table-variants.scss | scss/bootstrap/scss/mixins/ | - | 980.0 B |
| _text-truncate.scss | scss/bootstrap/scss/mixins/ | - | 168.0 B |
| _transition.scss | scss/bootstrap/scss/mixins/ | - | 661.0 B |

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https://ap-south-1.console.aws.amazon.com/s3/upload/cloudroute53?region=ap-south-1&bucketType=general

**Upload succeeded**
For more information, see the **Files and folders** table.

Files and folders | Configuration

**Files and folders** (139 total, 1.4 MB)

Find by name

1 2 3 4 5 6 7 ... 14

| Name | Folder | Type | Size | Status | Error |
|---|---|---|---|---|---|
| _grid.scss | scss/bootstrap/scss/mixins/ | - | 3.9 KB | Succeeded | - |
| _image.scss | scss/bootstrap/scss/mixins/ | - | 395.0 B | Succeeded | - |
| _list-group.scss | scss/bootstrap/scss/mixins/ | - | 509.0 B | Succeeded | - |
| _lists.scss | scss/bootstrap/scss/mixins/ | - | 168.0 B | Succeeded | - |
| _pagination.scss | scss/bootstrap/scss/mixins/ | - | 741.0 B | Succeeded | - |
| _reset-text.scss | scss/bootstrap/scss/mixins/ | - | 495.0 B | Succeeded | - |
| _resize.scss | scss/bootstrap/scss/mixins/ | - | 202.0 B | Succeeded | - |
| _table-variants.scss | scss/bootstrap/scss/mixins/ | - | 980.0 B | Succeeded | - |
| _text-truncate.scss | scss/bootstrap/scss/mixins/ | - | 168.0 B | Succeeded | - |
| _transition.scss | scss/bootstrap/scss/mixins/ | - | 661.0 B | Succeeded | - |

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### 3. Enable Static Website Hosting

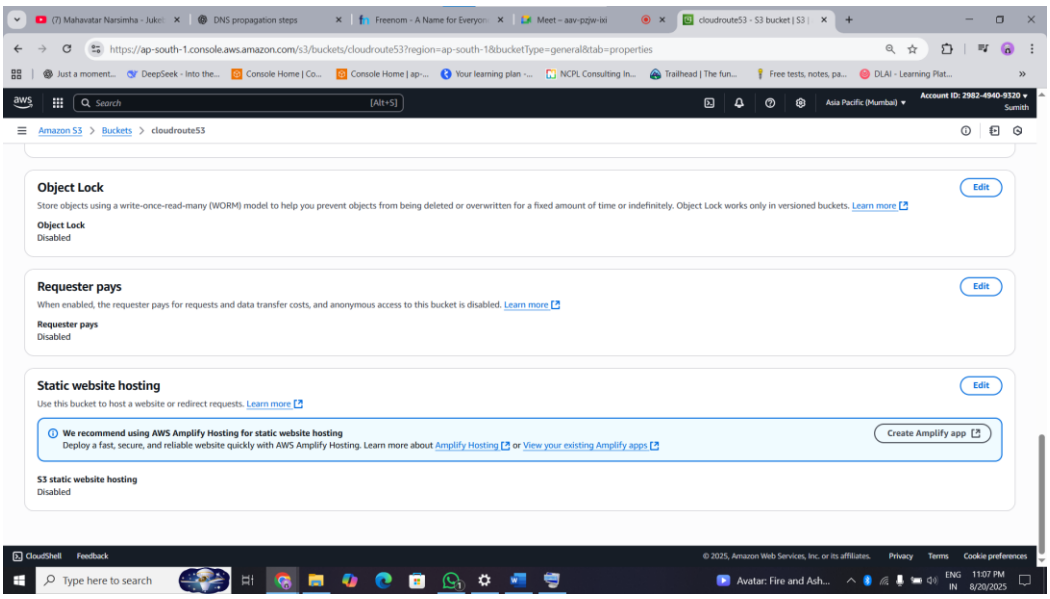
- Upload a simple index.html:

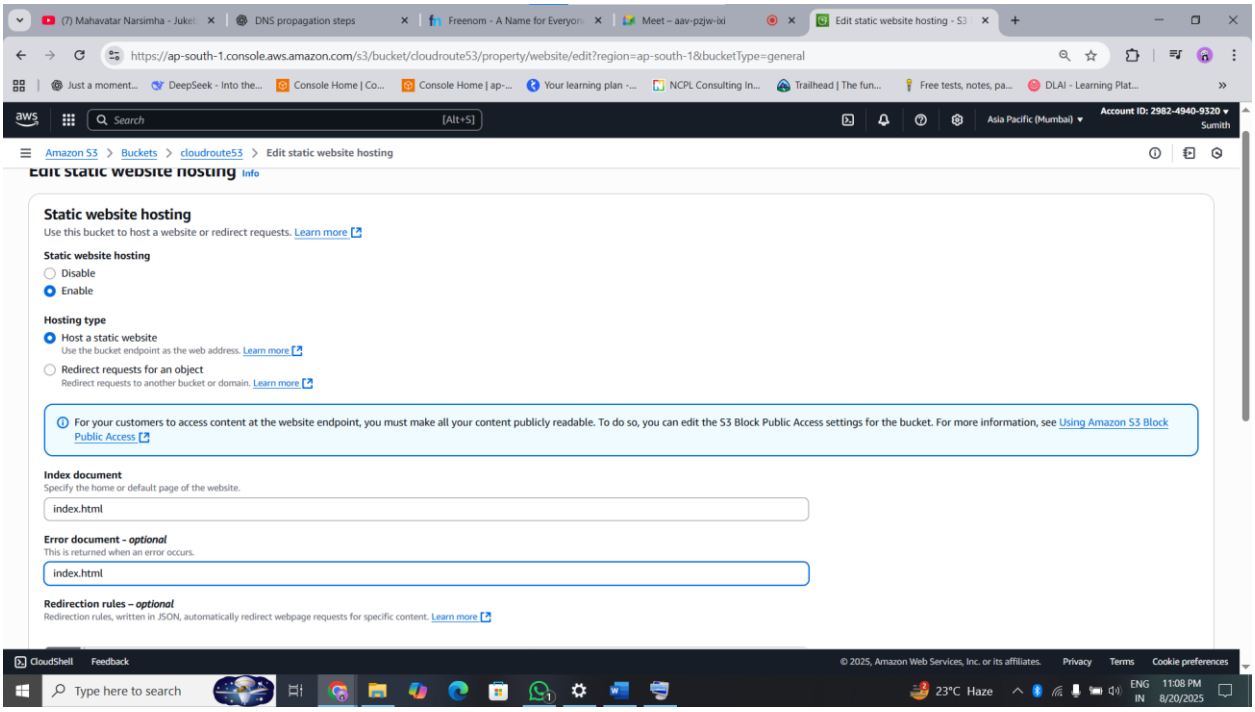
<html><body><h1>Hello from S3!</h1></body></html>

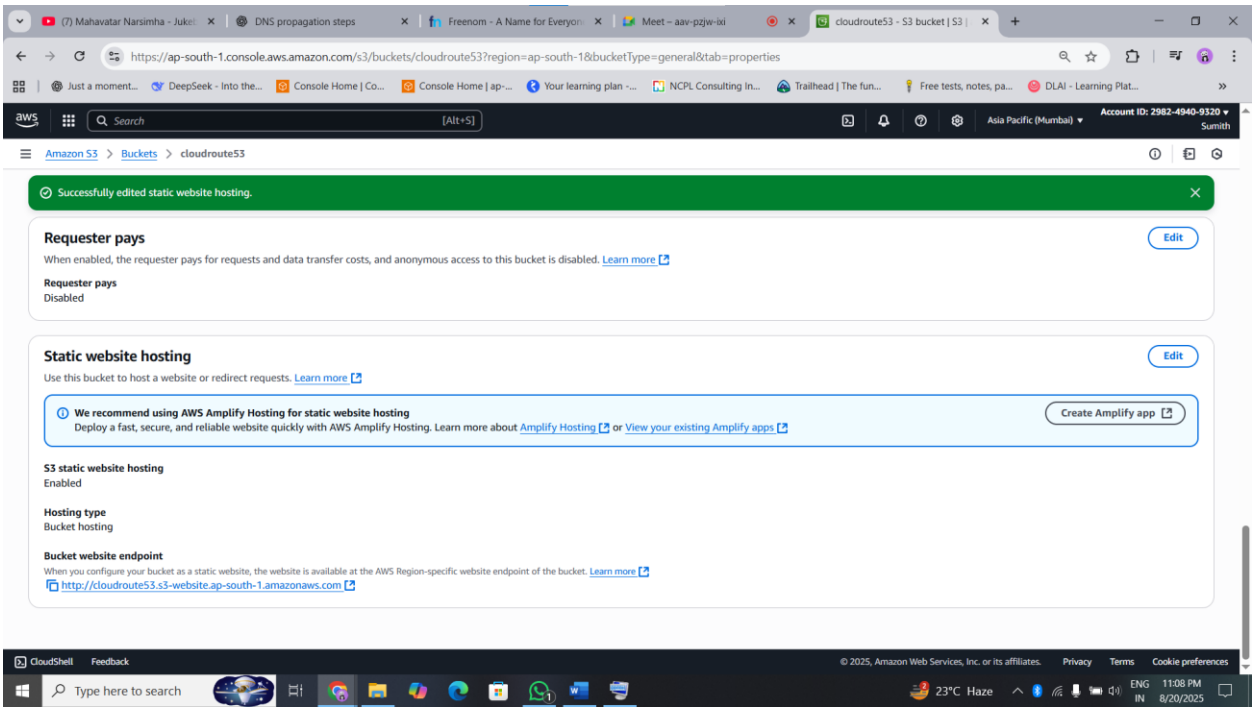
- Enable static website hosting:

aws s3 website s3://www.yourdomain.com/ \

 --index-document index.html --error-document error.html



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**4. Create a Hosted Zone in Route 53**

aws route53 create-hosted-zone \

--name yourdomain.com \

--caller-reference $(date +%s)

- Copy the nameservers (NS records) returned**.**

  ******NOTE: The below showcase the error on free domain.******

---

### 3. Update Domain Nameservers

- Since you are using **InfinityFree free domain (infy.uk subdomain)**, you **cannot change its nameservers** — InfinityFree controls it.
- ✅ If you had a custom domain (like `mydomain.com`), you'd go to your registrar and replace nameservers with the Route 53 ones.

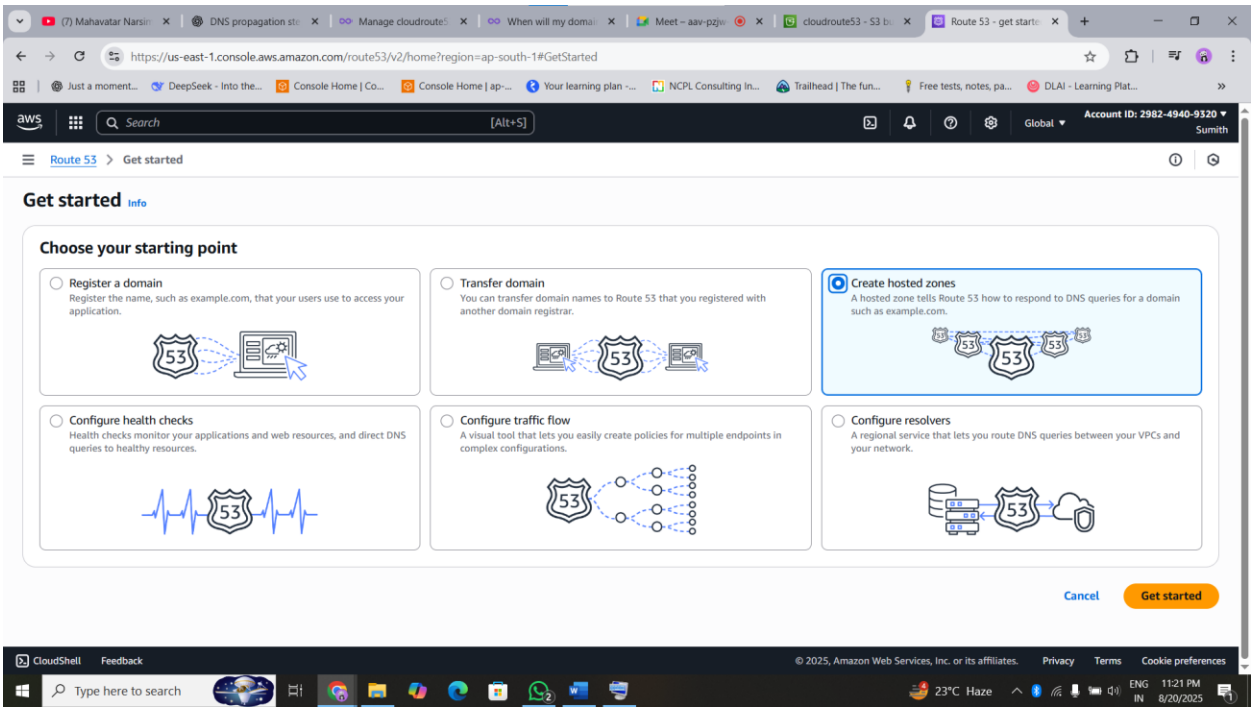
👉 But because this is a **InfinityFree free subdomain** (`infy.uk`), Route 53 cannot be the authoritative DNS. InfinityFree does not allow you to change NS for free domains.

---

### 4. Alternative (Workaround)

If you **really want to use Route 53**, you have 2 options:

1. **Buy a custom domain** (from Namecheap, GoDaddy, AWS, etc.), then update its NS to Route 53.
2. Keep using the InfinityFree subdomain (`cloudroute53.infy.uk`) → but Route 53 cannot control it. You'll only manage records inside InfinityFree DNS panel.


https://us-east-1.console.aws.amazon.com/route53/v2/home?region=ap-south-1#GetStarted
Account ID: 2982-4940-9320
Sumith
Route 53 > Get started
Get started Info
Choose your starting point
Register a domain
Register the name, such as example.com, that your users use to access your application.
Transfer domain
You can transfer domain names to Route 53 that you registered with another domain registrar.
Create hosted zones
A hosted zone tells Route 53 how to respond to DNS queries for a domain such as example.com.
Configure health checks
Health checks monitor your applications and web resources, and direct DNS queries to healthy resources.
Configure traffic flow
A visual tool that lets you easily create policies for multiple endpoints in complex configurations.
Configure resolvers
A regional service that lets you route DNS queries between your VPCs and your network.
Cancel
Get started
CloudShell
Feedback
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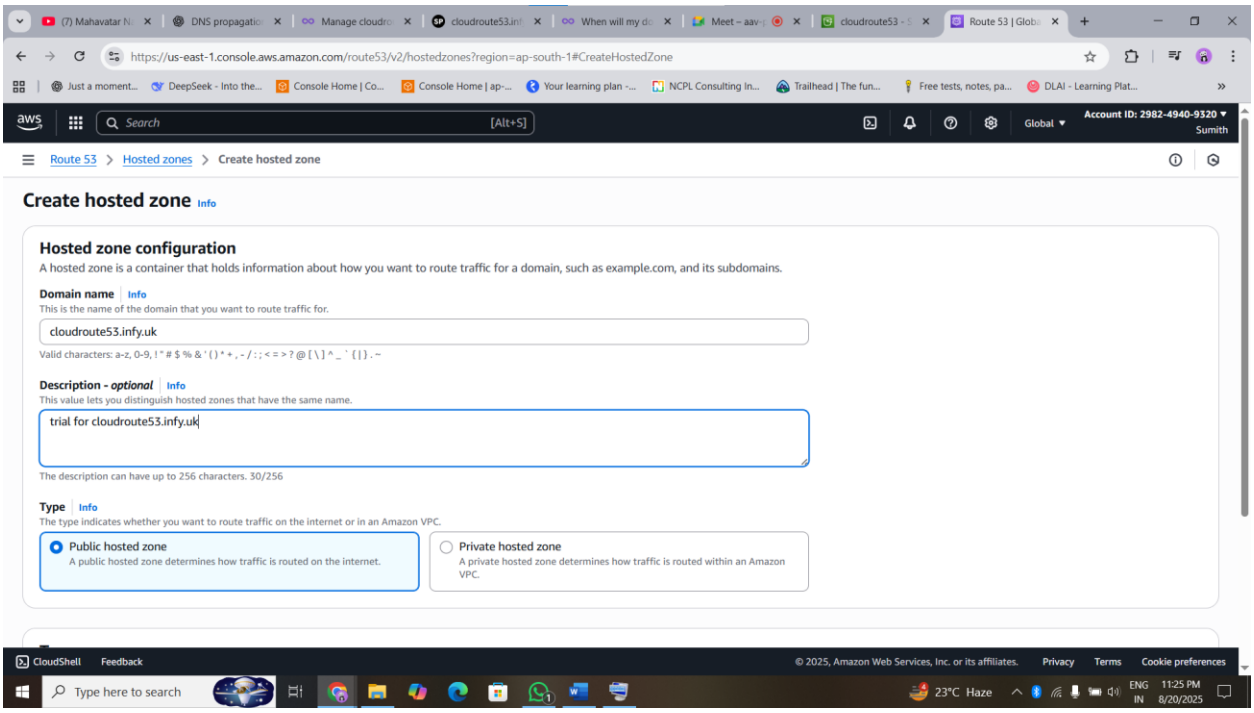
Privacy
Terms
Cookie preferences


https://us-east-1.console.aws.amazon.com/route53/v2/hostedzones?region=ap-south-1#CreateHostedZone
Account ID: 2982-4940-9320
Sumith
Route 53 > Hosted zones > Create hosted zone
Create hosted zone Info
Hosted zone configuration
A hosted zone is a container that holds information about how you want to route traffic for a domain, such as example.com, and its subdomains.
Domain name Info
This is the name of the domain that you want to route traffic for.
cloudroute53.infy.uk
Valid characters: a-z, 0-9, ! " # $ % & ' ( ) * + , - / : ; < = > ? @ [ \ ] ^ _ ` { | } . ~
Description - optional Info
This value lets you distinguish hosted zones that have the same name.
trial for cloudroute53.infy.uk
The description can have up to 256 characters. 30/256
Type Info
The type indicates whether you want to route traffic on the internet or in an Amazon VPC.
Public hosted zone
A public hosted zone determines how traffic is routed on the internet.
Private hosted zone
A private hosted zone determines how traffic is routed within an Amazon VPC.
CloudShell
Feedback
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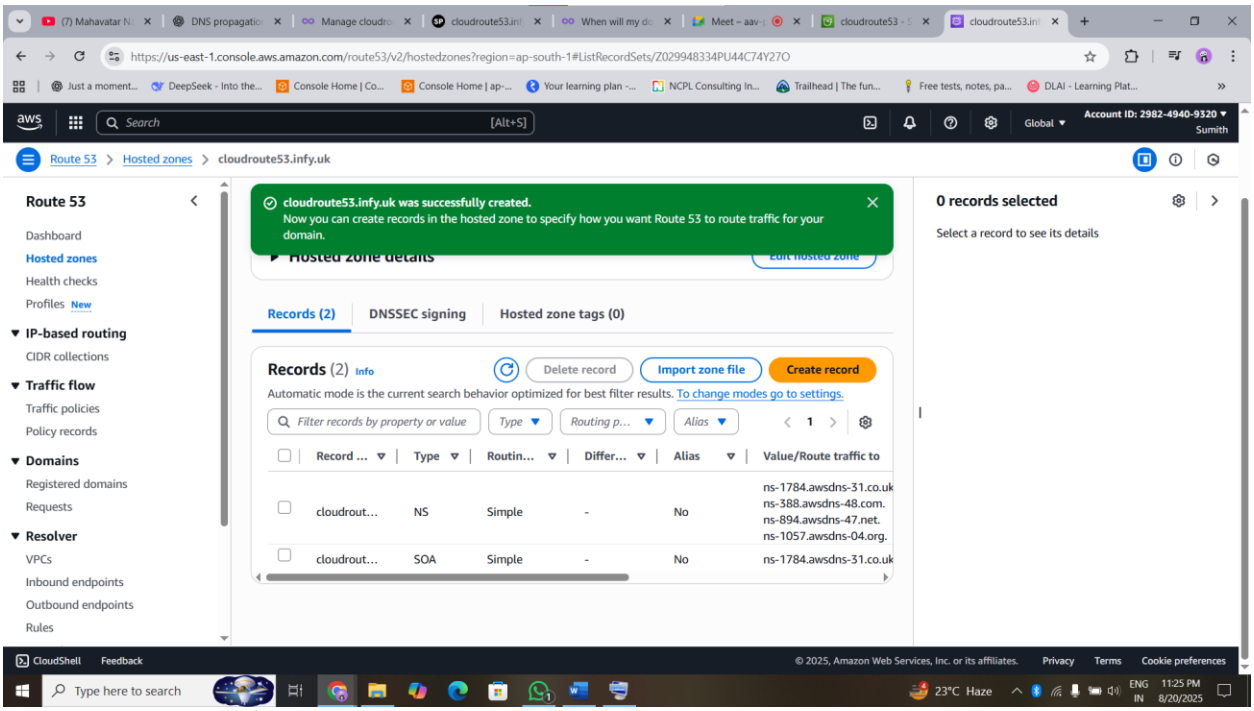
Privacy
Terms
Cookie preferences

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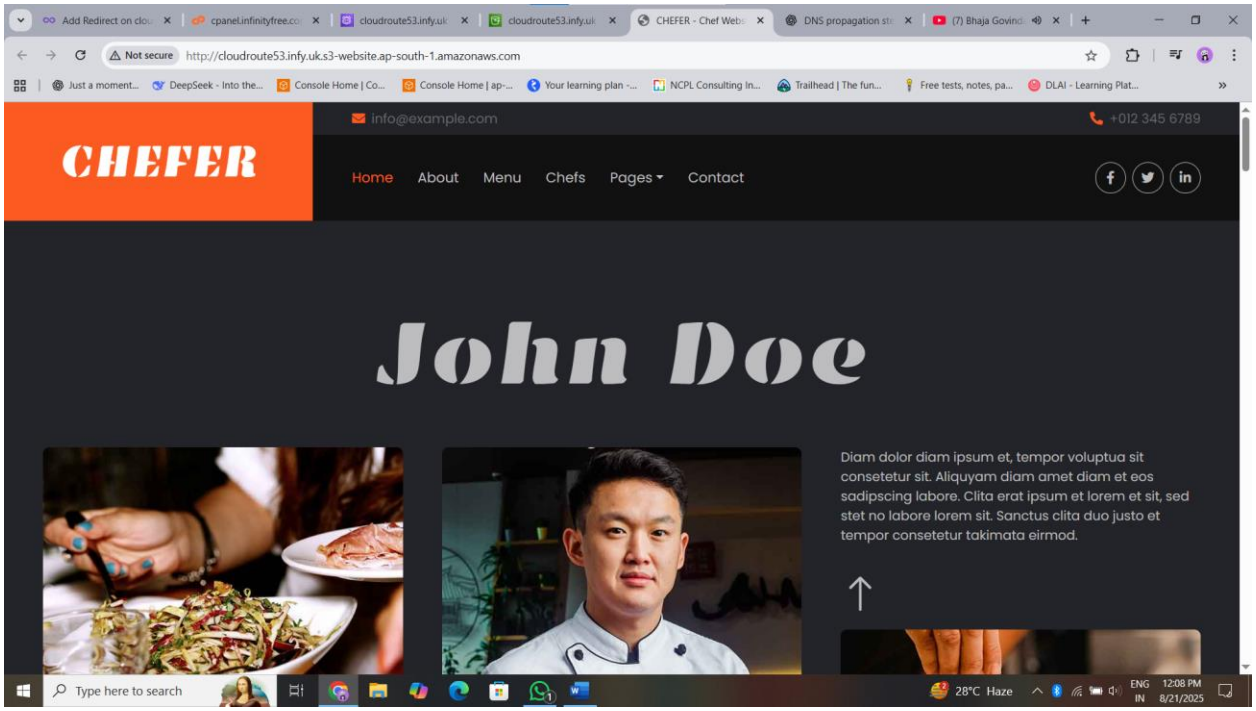
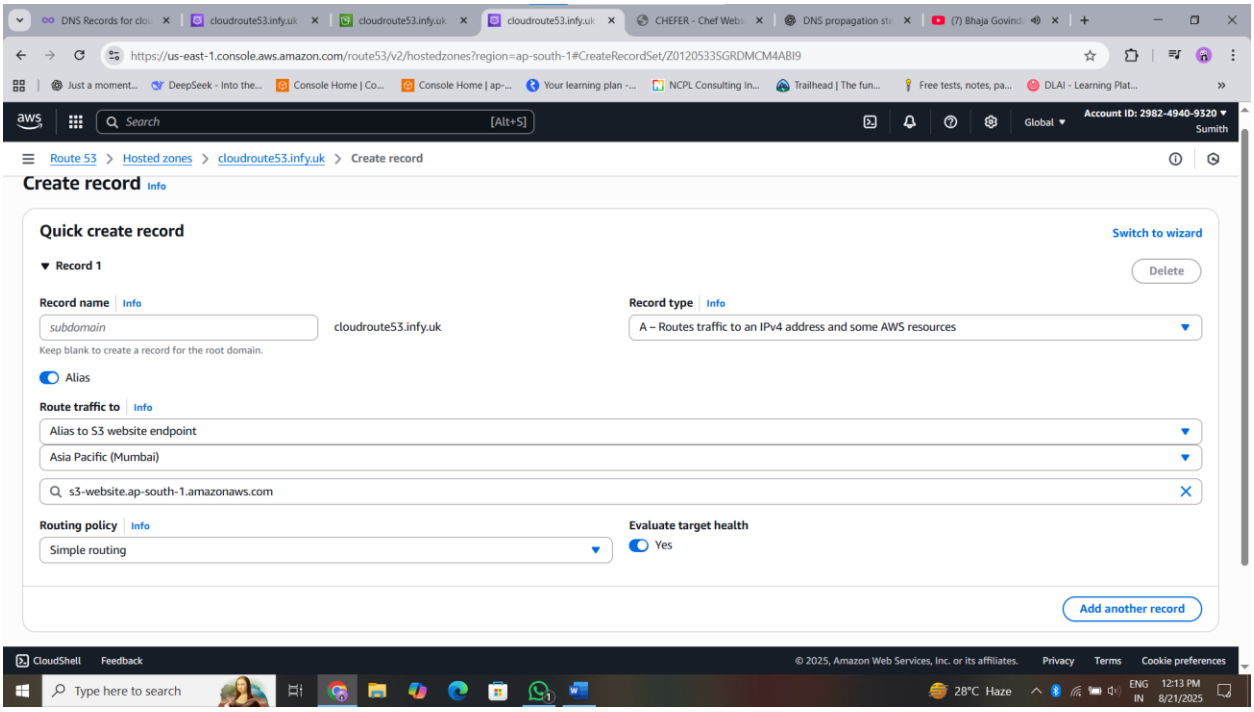
## 5. Update Nameservers at Registrar

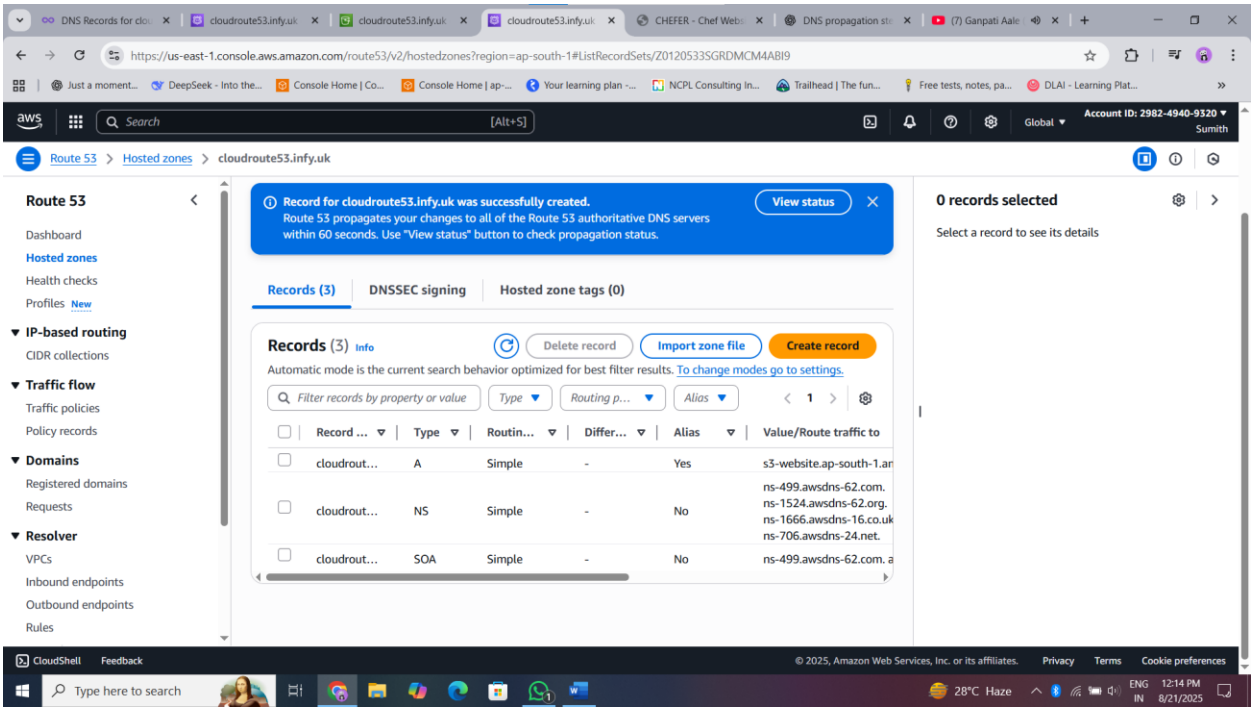
- Go to your registrar (GoDaddy, Namecheap, etc.).
- Replace existing nameservers with the Route 53 NS records.
- Wait for propagation (~10–30 min, sometimes up to 24 hrs**).**



## 6. Create an A-Record Alias for S3 Website Endpoint

- Go to Route 53 → Hosted Zone → Add Record.
- Record type: A – IPv4
- Alias: Yes → Point to your S3 static website endpoint (not the bucket ARN).



**Verify:**

👉 **Open http://www.yourdomain.com → should load your S3 site.**

---

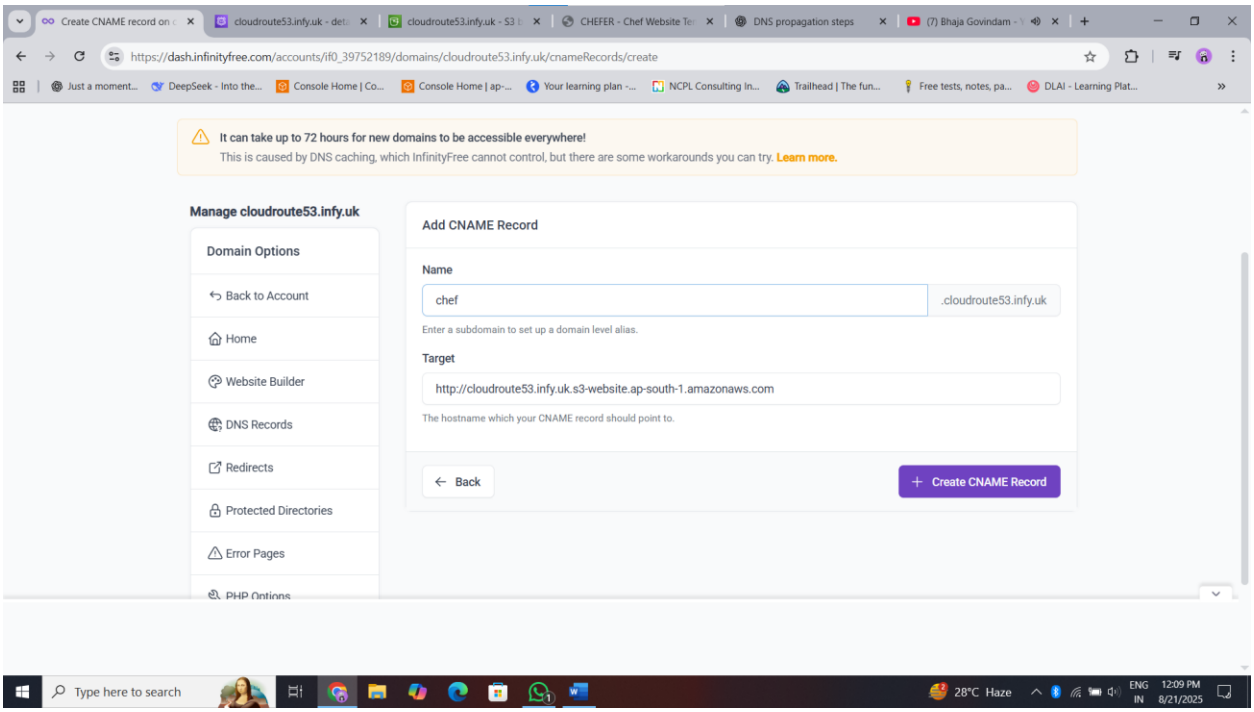
**7. Request SSL Certificate from ACM**

```
aws acm request-certificate \
  --domain-name www.yourdomain.com \
  --validation-method DNS \
  --region us-east-1
```

---

**8. Validate the Certificate**

- ACM will give you a CNAME record.
- Add it to your Route 53 hosted zone.
- Wait for ACM to mark the certificate as Issued**.**




---

**9. Create CloudFront Distribution**

- Origin: your S3 Website Endpoint (⚠️ not the bucket directly).
- Viewer Protocol Policy: Redirect HTTP → HTTPS.

- Associate the ACM certificate for www.yourdomain.com.

---

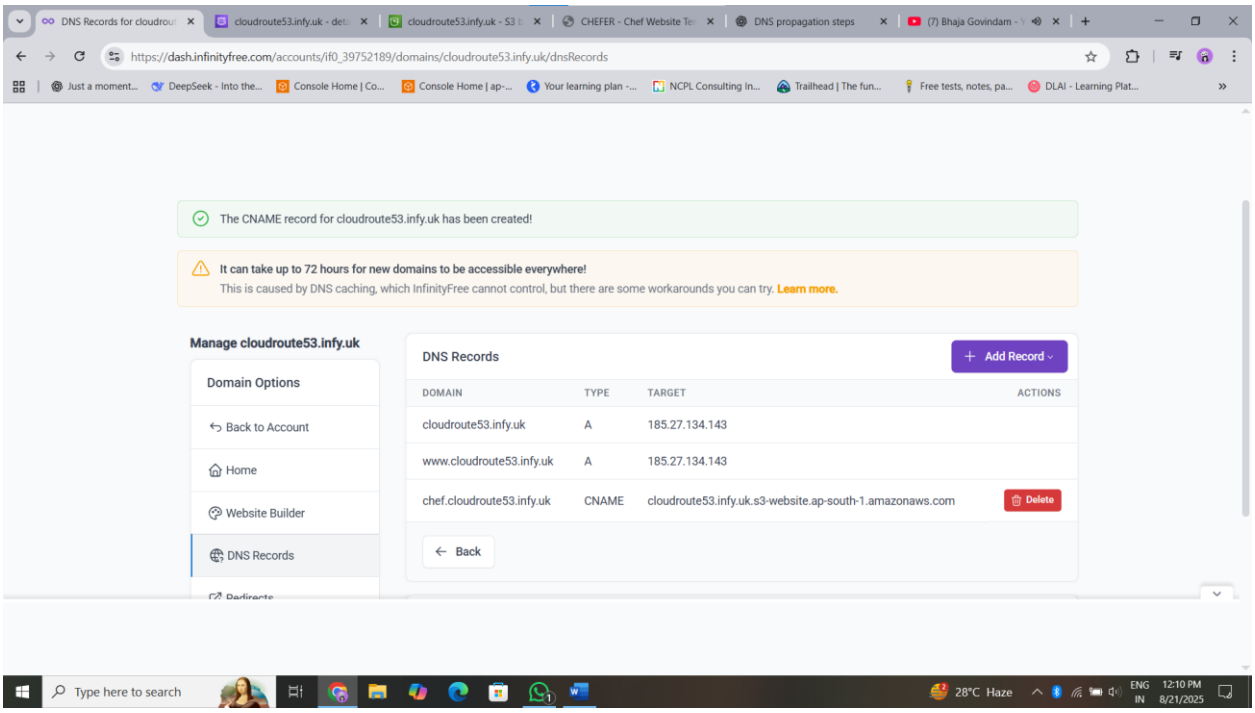
### 10. Update Route 53 A-Record to CloudFront

- Change your A-record to alias → CloudFront distribution domain (e.g., d12345.cloudfront.net).

---

### 11. Test Your Setup

- Wait for DNS propagation.
- Visit:
  - ✅ http://www.yourdomain.com → redirects to HTTPS
  - ✅ https://www.yourdomain.com → loads securely via CloudFront



**Task 2.1: Understand DNS Propagation**

🔷 **Objective**

To learn how DNS propagation works and why DNS changes don't reflect immediately worldwide.

---

🔷 **Steps**

1. **Make a DNS Change in Route 53**
   - Examples:
     - Update a domain's **nameservers**.
     - Modify an **A record** (point it to a new IP).
     - Update a **CNAME record** for a subdomain.

👉 Remember: TTL (Time to Live) settings in your record define how long DNS resolvers cache the old value.

2. **Wait and Check Propagation**
   - Changes don't reflect instantly; they can take from a few minutes up to **48 hours** depending on the record type, TTL, and ISP caching.
   - Propagation speed also differs based on **geographic region**.
3. **Verify Using Tools**
   - Use DNSChecker or WhatsMyDNS.
   - Enter your domain and select the record type (A, CNAME, MX, etc.).
   - The tool will show results from DNS servers around the world.
4. **Observe the Propagation Pattern**
   - You'll see some servers showing the **old value** and others showing the **new value** until propagation completes.
   - This demonstrates real-world delays and why troubleshooting DNS requires patience.

---

🔷 **Critical Troubleshooting Insight for Production**

- **Users in one region** may see the new website, while others still see the old one.
- Always **plan DNS changes** (especially nameserver updates) during low-traffic windows.
- Reduce **TTL values** to speed up propagation **before** making a major change.
- Always verify changes using global tools, not just your local computer.

**Task 2.2: Break and Fix Permissions**

🔷 **Objective**

To understand how **S3 bucket policies** and **IAM permissions** affect access, and why restricting direct S3 access is a best practice when using **CloudFront**.

---

🔷 **Steps**

**1. Verify Current State**

- Your website is already hosted on S3 and served via **CloudFront**.
- At this point:
  - ✅ Accessing the **S3 static website endpoint** (like http://mybucket.s3-website-us-east-1.amazonaws.com) should work.
  - ✅ Accessing via **CloudFront distribution domain** (like https://d12345.cloudfront.net) should also work.

---

**2. Introduce Restriction (Break Access)**

- Create a **CloudFront Origin Access Identity (OAI)** if you don't already have one.
  - This acts as a "virtual user" that CloudFront uses to fetch objects from your S3 bucket securely.
- Update the **S3 bucket policy** to **allow only CloudFront's OAI** to access it.

  Example bucket policy (replace with your bucket ARN and OAI):

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AllowCloudFrontServicePrincipal",
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam::cloudfront:user/CloudFront Origin Access Identity E3EXAMPLEOAI"
      },
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::mybucket/*"
```

```
  }
 ]
}
```

- Remove any "Principal": "*" public access statements that previously allowed everyone to read objects.

---

**3. Test the Break**

- Now try accessing your **S3 website endpoint directly**.
  - ❌ It should fail (Access Denied).
- Try accessing your site via **CloudFront distribution URL**.
  - ✅ It should still work fine, since CloudFront (via OAI) has permission.

---

**4. Fix and Validate**

- If something breaks unexpectedly, check:
  - Bucket policy (correct OAI ARN used).
  - CloudFront distribution → Origins → “Restrict Bucket Access” setting.
  - Whether you’re using **S3 static website endpoint** vs **S3 REST API endpoint** (OAI works with the REST API endpoint, not the website endpoint).
- Once fixed:
  - Direct S3 access stays blocked.
  - CloudFront access continues to work.

---

🔷 **Critical Production Insight**

- **Best practice**: Always serve public content via **CloudFront** instead of exposing S3 buckets directly.
- Benefits:
  - 🔒 Security: S3 stays private.
  - 🚀 Performance: CloudFront caches content globally.
  - 💰 Cost: Reduces S3 direct requests and data transfer costs.

**Task 2.3: Test the CDN Performance**

🔷 **Objective**

**To visually experience the benefit of CloudFront (CDN) in accelerating content delivery across the globe compared to accessing directly from S3.**

---

🔷 **Steps**

**1. Prepare the Test File**

- **Upload a large image or file (e.g., 5–10 MB PNG/JPG or a video snippet) into your S3 bucket.**
- **Make sure it's accessible via your CloudFront distribution.**
- **Example URLs:**
  - **S3 direct: https://mybucket.s3.amazonaws.com/largeimage.jpg**
  - **CloudFront: https://d1234abcdef.cloudfront.net/largeimage.jpg**

---

**2. Link File in Your Webpage**

- **Edit your website HTML to include the image (served through CloudFront):**

```
<img src="https://d1234abcdef.cloudfront.net/largeimage.jpg" alt="CDN Test Image" width="600">
```

---

**3. Run Performance Tests**

- **Use tools like:**
  - **Pingdom**
  - **GTmetrix**
  - **WebPageTest**
- **Run the test from different geographic locations (e.g., US, Europe, Asia).**
- **Record two sets of tests:**

1. **File served directly from S3 (no CDN).**
2. **File served from CloudFront (via edge locations).**

---

**4. Observe and Compare**

- **Without CDN (S3 only):**
  - **All requests must travel to the S3 bucket's region (e.g., us-east-1).**

    - **Higher latency for distant regions.**
- **With CDN (CloudFront):**
    - **Requests are routed to the nearest CloudFront edge location.**
    - **Faster load times, reduced latency, and cached responses after first request.**

---

🔷 **Critical Insight for Production**

- **CDNs don't just improve speed — they also:**
    - ✅ **Reduce origin load (fewer direct requests to S3).**
    - ✅ **Improve global availability.**
    - ✅ **Lower latency spikes during traffic surges.**

**Task 3.1: Implement a Blue/Green Style Deployment with Route 53**

:

---

🔷 **Objective**

To simulate a **Blue/Green (or Canary) deployment** using **Amazon Route 53's Weighted Routing Policy**, allowing you to gradually shift traffic between two environments.

---

🔷 **Steps**

**1. Create the Blue and Green Environments**

- Create **two S3 buckets**:
    - blue-website.yourdomain.com
    - green-website.yourdomain.com
- Enable **Static Website Hosting** for both buckets.
    - Upload an index.html file with a **blue background** for the blue site.
    - Upload an index.html file with a **green background** for the green site.
    - Example:
        - Blue version: <body style="background-color:blue; color:white;">Blue Website</body>
        - Green version: <body style="background-color:green; color:white;">Green Website</body>

---

**2. Set Up Route 53 Weighted Routing**

- Go to **Route 53 → Hosted Zones** → Select your hosted zone (for yourdomain.com).
- Create a **new record set**:

**Record 1 (Blue)**

- **Name**: test.yourdomain.com
- **Type**: A – IPv4 address (Alias)
- **Alias Target**: S3 website endpoint for Blue bucket
- **Routing Policy**: Weighted
- **Weight**: 90
- **Set ID**: Blue

**Record 2 (Green)**

- Same as above, but:
  - Alias Target = Green bucket's S3 website endpoint
  - Weight = 10
  - Set ID = Green

---

**3. Test the Deployment**

- Open your browser and repeatedly visit:

  👉 http://test.yourdomain.com
- Expected behavior:
  - **90% of the time** → Blue site (blue background).
  - **10% of the time** → Green site (green background).

This simulates a **canary release** where most users stay on the stable version (Blue), while a small fraction test the new version (Green).

---

**4. Shift Traffic**

- Later, you can **adjust the weights** in Route 53:
  - Move to 70/30, 50/50, and eventually 0/100 (all traffic on Green).
- This demonstrates **Blue/Green cutover with no downtime**.

---

🔷 **Critical Production Insight**

- **Canary deployment** lets you test new features with a small subset of users before full rollout.
- **Blue/Green deployments** reduce risk by keeping two parallel environments.
- **Route 53 weighted routing** is often used for **gradual migrations** and **A/B testing**.

**Task 3.2: Automate the Deployment with AWS CL**

🔷 **Steps to Automate Blue/Green Deployment with AWS CLI**

**1. Create the S3 Buckets**

```
aws s3 mb s3://blue-website.yourdomain.com --region us-east-1
aws s3 mb s3://green-website.yourdomain.com --region us-east-1
```

---

**2. Enable Static Website Hosting**

For Blue bucket:

```
aws s3 website s3://blue-website.yourdomain.com/ \
  --index-document index.html --error-document error.html
```

For Green bucket:

```
aws s3 website s3://green-website.yourdomain.com/ \
  --index-document index.html --error-document error.html
```

---

**3. Upload Files to Buckets**

```
aws s3 sync ./site-blue s3://blue-website.yourdomain.com
aws s3 sync ./site-green s3://green-website.yourdomain.com
```

*(Here ./site-blue and ./site-green are your local folders with HTML files.)*

---

**4. Find Your Hosted Zone ID**

```
aws route53 list-hosted-zones-by-name \
  --dns-name yourdomain.com \
  --query "HostedZones[0].Id" --output text
```

Save the ZONE_ID returned.

---

**5. Create Weighted Route 53 Records**

Create a JSON file weighted-records.json:

```
{
  "Comment": "Weighted routing for Blue/Green deployment",
  "Changes": [
    {
      "Action": "CREATE",
      "ResourceRecordSet": {
        "Name": "test.yourdomain.com",
        "Type": "A",
        "SetIdentifier": "Blue",
        "Weight": 90,
        "AliasTarget": {
          "HostedZoneId": "Z3AQBSTGFYJSTF",
          "DNSName": "s3-website-us-east-1.amazonaws.com.",
          "EvaluateTargetHealth": false
        }
      }
    },
    {
      "Action": "CREATE",
      "ResourceRecordSet": {
        "Name": "test.yourdomain.com",
        "Type": "A",
        "SetIdentifier": "Green",
        "Weight": 10,
        "AliasTarget": {
          "HostedZoneId": "Z3AQBSTGFYJSTF",
          "DNSName": "s3-website-us-east-1.amazonaws.com.",
          "EvaluateTargetHealth": false
        }
      }
```

```
    }
  ]
}
```

Apply the change:

```
aws route53 change-resource-record-sets \
  --hosted-zone-id ZONE_ID \
  --change-batch file://weighted-records.json
```

---

**6. (Advanced) Create CloudFront Distribution**

```
aws cloudfront create-distribution \
  --origin-domain-name blue-website.yourdomain.com.s3.amazonaws.com
```

- Get the CloudFront domain name from the output.
- Update your Route 53 record to point to CloudFront instead of S3 directly.

---

**7. Test Deployment**

- Visit http://test.yourdomain.com multiple times.
- You should see Blue site 90% of the time and Green site 10% of the time.